F1 Strategic Intelligence:

Optimization of Race Strategy –

Australian GP 2026

Index

1. Context & Race Overview *3*

2. Strategic Inflection Point *4*

3. Performance & Tyre Degradation *5*

3.1 Data filtering methodology **5**

3.2 Analysis of stint 1: Car #16 (Leclerc) on Medium C4 ..... **6**

3.3 Analysis of stint 2: Car #63 (Russell) on Hard C3 ..... **6**

3.4 Strategic interpretation **6**

4. Simulation of the "Optimal Strategy" *7*

5. Conclusions & Key Learnings *8*

6. Appendix *9*

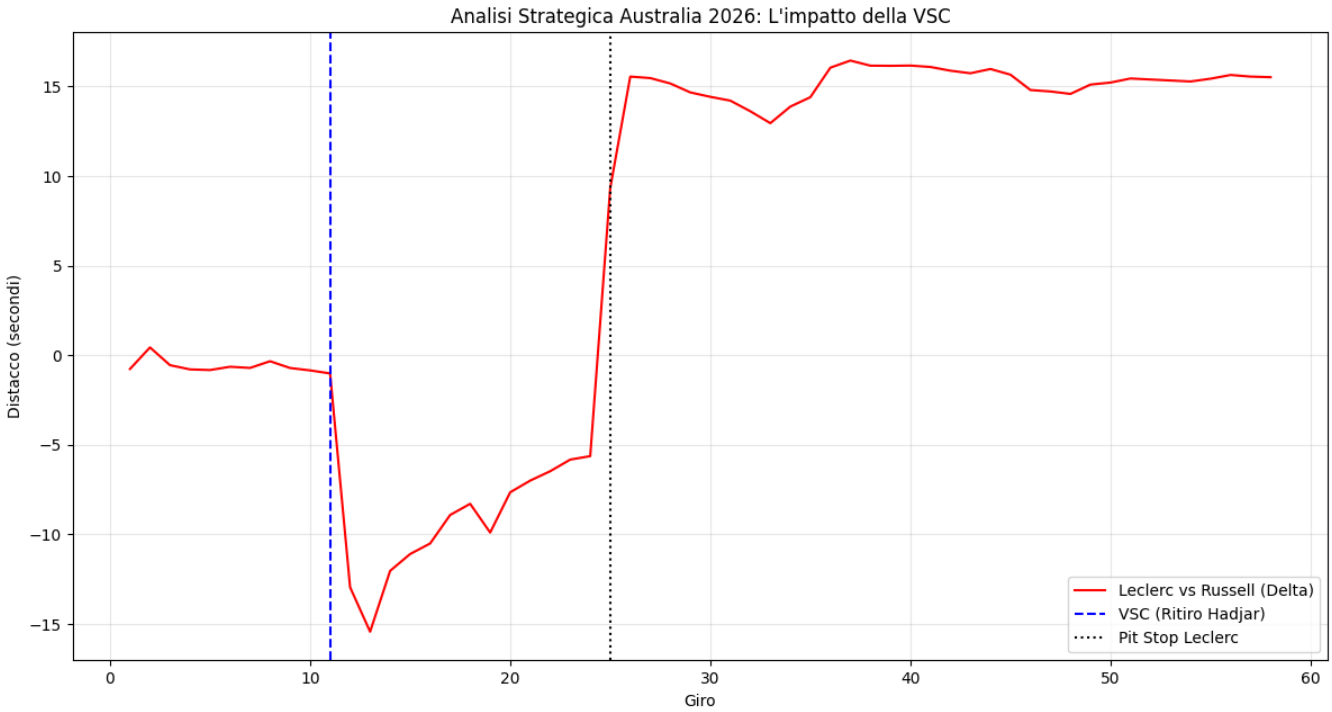
1. Context & Race Overview

The analysis in question concerns the first race of 2026, namely the race held in Australia, in Melbourne; in particular, I will take care of the analysis of the race by observing the #16 car (Charles Leclerc), driver of Scuderia Ferrari.

The race began with a great start from Leclerc, immediately moving into pole position and maintaining this advantage over the course of the race. On lap 11, Leclerc had a 0.8-second advantage over the #63 car (George Russell), a Mercedes driver; both cars were fitted with a Medium C4 compound from the start of the race, with the aim of making a single stop and then fitting a Hard C3 compound. Also on lap eleven, Hadjar, a Red Bull driver, had a problem with his car, forcing him to retire and the consequent yellow flag, entering the Virtual Safety Car (VSC) regime.

The VSC lasted from lap 11 to lap 14 and during this period both Mercedes cars opted for a pitstop on a hard compound, which will take them to the end of the race without making any further pit stops. Scuderia Ferrari decided not to stop either car, presumably remaining on the initial strategy.

2. Strategic Inflection Point



Delta's Time Cumulative analysis shows that the decision not to react to Russell's stop on lap 11 created a "false advantage". While Leclerc was actually leading the race, he was racking up a strategic debt of 10.7 seconds (VSC Gain lost) which proved fatal at the standard stop on lap 25.

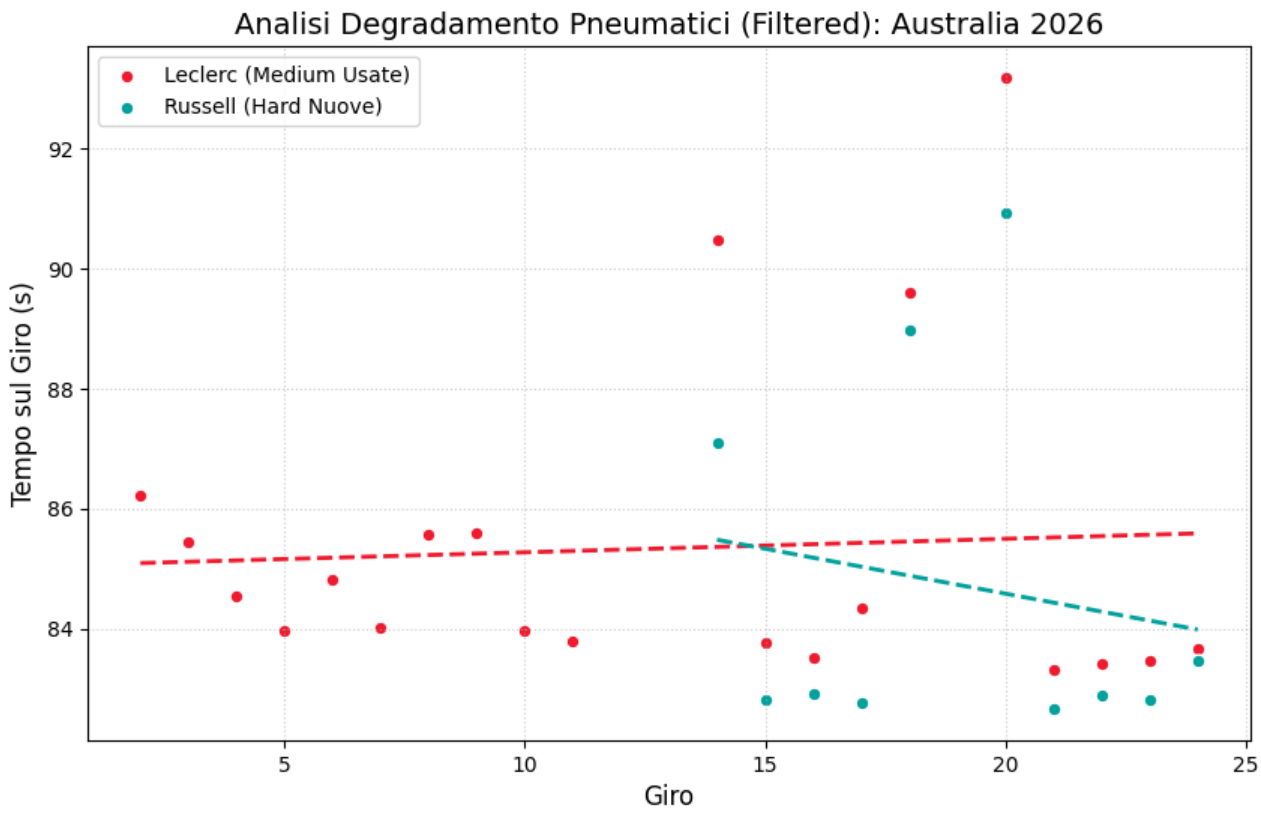
3. Performance & Tyre Degradation

3.1 Data filtering methodology

To ensure an accurate analysis and a correct reading of the race pace, the raw data has been subjected to a Data Cleaning process. Performance outliers above 107% of the average lap time, which brought the graphical representation of the data "off course", have been removed. The exclusions made were:

- Giro 11: regime di Virtual Safety Car
- Pit entry and exit laps
- Heavy traffic phases

This was done to isolate the pure performance decay due solely to the wear of the fitted compound and fuel consumption.



3.2 Analysis of stint 1: Car #16 (Leclerc) on Medium C4

The analysis shows excellent management of the compound by Leclerc, but limited by the cliff performance of the compound.

- Coefficient of Degradation: The value of +0.0225 s/rev indicates extremely low degradation (almost zero). This shows that Leclerc was managing the tyre perfectly, keeping the Medium compound temperatures under control despite the prolonged stint.
- Compound Limitations: Despite the stability of degradation, the performance floor of the used tyre (the minimum achievable time) was now too high compared to cars with new tyres.

3.3 Analysis of stint 2: Car #63 (Russell) on Hard C3

The direct comparison with George Russell (Mercedes) highlights the real damage suffered by Ferrari.

- Russell Degradation Coefficient: The value of -0.1496 s/rev (negative) is impressive. It indicates that Russell was not degrading, but was improving his times as the car became lighter for fuel consumption. The new Hard tyre was in its ideal shape.
- Pace Delta Medium: The average pitch differential was 0.601 seconds per lap in favor of Mercedes.
- Accumulated Damage: In the 14 laps between Russell's stop (Lap 11) and Leclerc's (Lap 25), the Ferrari #16 accumulated a "passive" delay of about 8.4 seconds (0.601s * 14 laps) only due to the difference in pure performance between the compounds.

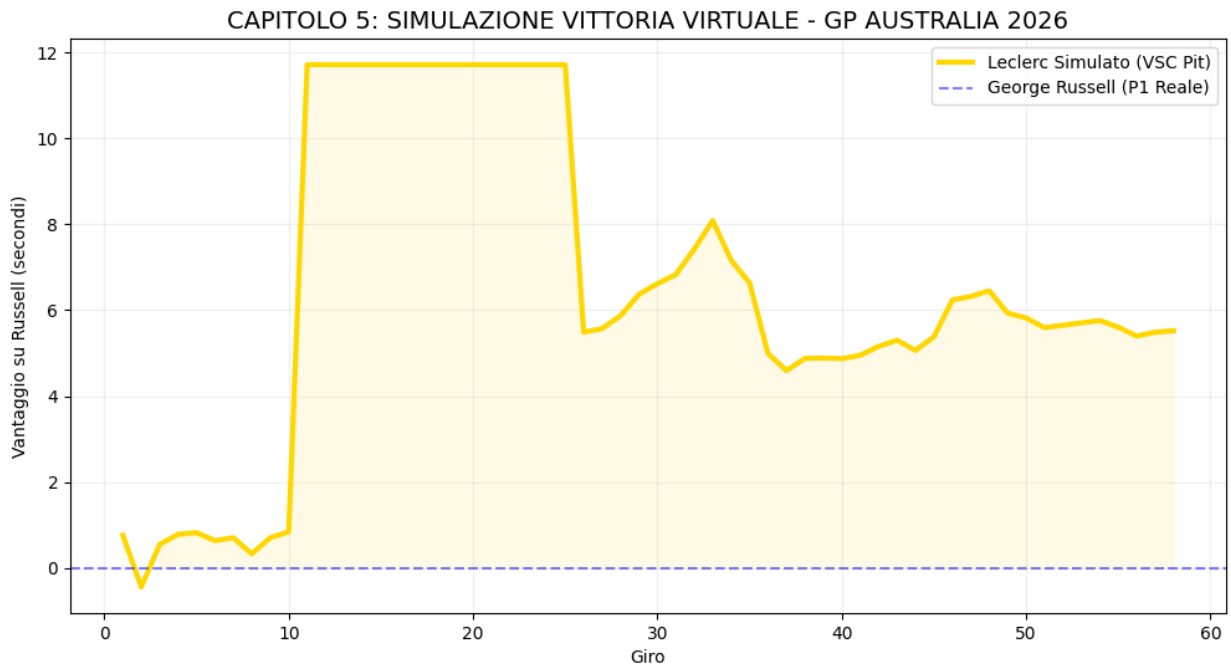
3.4 Strategic interpretation

We learned that Leclerc was performing excellent tyre management with limited degradation at 0.0225 sec/lap, but Ferrari's choice to extend the stint on Medium exposed Leclerc to a negative Pace Delta of 0.601 sec/lap. Adding the loss of chronometric performance (8.4s) to the loss of the pit stop gain under VSC (estimated at ~10s), the overall strategic loss before Leclerc's pit stop amounts to about 18.4 seconds.

4. Simulation of the "Optimal Strategy"

Assuming the adoption of a reactive strategy (Covering Strategy) in response to Mercedes' manoeuvre on lap 11, the race of the #16 car, i.e. Leclerc's, would have ended differently and, above all, with a more than positive outcome.

The software recalculated Leclerc's race history, integrating the time advantage of the pit stop under VSC (VSC Gain) and aligning his race pace with that of his competitors on the Hard C3 compound. Entering the pits on lap 11 would have guaranteed a net saving of 10.7 seconds compared to the green flag stop on lap 25, thus neutralizing the strategic debt. Furthermore, by adopting the Hard C3 compound in the second stint, Leclerc would have cancelled out the negative Pace Delta of 0.601 seconds per lap suffered in reality due to wear on the Medium compound, thus recovering the performance gap. The simulation shows that Charles Leclerc would have maintained the race lead after the pit stop sequence, crossing the finish line with a simulated margin of victory of 5.52 seconds over George Russell.



5. Conclusions & Key Learnings

The technical analysis of the 2026 Australian Grand Prix offers valuable insights into the management of race dynamics and the responsiveness of the pit wall. Charles Leclerc showed excellent compound management, with minimal degradation of just +0.0225 seconds per lap. Unfortunately, his pure performance was compromised by a strategic positioning error. The team underestimated the crossover point, extending the stint too much on the Medium and ignoring the performance advantage of the new Hard compound, which even had a negative degradation of -0.1496 seconds per lap on Russell's car. The Virtual Safety Car on lap 11 was the only real strategic turning point of the race. Not using it to stop turned a real advantage into a passive delay of 18.4 seconds before the actual pit stop. The lesson learned is clear: in the presence of neutralizations (VSC/SC) and with a reduced advantage (less than a second), the top priority must be the protection of the track position with an "economical" stop, especially if the direct opponents aim for the undercut.

6. Appendix

The entire analysis has been developed through an advanced programming ecosystem to ensure maximum accuracy of the extracted data.

- Data Sourcing: Extraction of official telemetry data and lap times via the FastF1 library in Python 3.14 environment.
- Data Cleaning & Filtering: Implementation of a filtering algorithm for the automatic removal of performance outliers (VSC, pit entry/exit laps and heavy traffic), isolating laps with times 107% higher than average.
- Degradation Modeling: Use of linear regression algorithms (NumPy library) for the calculation of performance decay coefficients. A degradation of +0.0225 s/lap for the Medium compound and an improvement of -0.1496 sec/lap for the Hard compound has been isolated.
- Simulation Engine: Development of a proprietary race simulator for the reconstruction of cumulative times. The model integrates the Pace Delta of 0.601s and the VSC Gain time advantage of 10.7s to generate the optimal strategic scenario.
- Data Visualization: Generation of professional high-resolution graphs using Matplotlib for the representation of time gaps and degradation curves.